

Homework # 7: Monte Carlo Methods

Due April 11, printed or emailed before 5:30 pm

Solve #1-2 by hand.

- Let U be a Standard Uniform random variable. Derive a formula that will convert U into
 - an Exponential random variable with the parameter $\lambda = 2.5$;
 - a Bernoulli random variable with the probability of success 0.77;
 - a discrete random variable with the distribution $P(x)$, where $P(0) = 0.2$, $P(2) = 0.4$, $P(7) = 0.3$, $P(11) = 0.1$;
 - a continuous random variable with the density $f(x) = 3x^2$ and c.d.f. $F(x) = x^3$, $0 < x < 1$;
 - a continuous random variable with the density $f(x) = 1.5x^2$ and c.d.f. $F(x) = (x^3 + 1)/2$, $-1 < x < 1$.

Our computer generates Uniform random variables,

0.119, 0.234, 0.604, 0.434, 0.694, 0.009, 0.772, 0.655, 0.078, 0.003

Use them to compute the variables generated according to your algorithms in (a-e).

- Two mechanics are changing oil filters for the arrived customers. The service time has an Exponential distribution with the parameter $\lambda = 5 \text{ hrs}^{-1}$ for the first mechanic, and $\lambda = 20 \text{ hrs}^{-1}$ for the second mechanic. Since the second mechanic works faster, he is serving 4 times more customers than his partner. Therefore, when you arrive to have your oil filter changed, your probability of being served by the faster mechanic is $4/5$. Let X be your service time. Explain how to generate the random variable X .

Solve our old homework problems on WinBUGS or R.

- (HW-3, #1) Service times of a queuing system follow Exponential distribution with an unknown parameter θ . Assume a Gamma(α, λ) prior distribution with $\alpha = 4$ and $\lambda = 2$. (Notice that WinBUGS and R understand the second parameter of `dgamma(alpha,lambda)` as the “rate” parameter λ , so that the mean of this distribution is α/λ .)

If a sample $\mathbf{X} = (0.7, 0.2, 2.5, 3.7, 0.9)$ is observed, find the posterior mean and variance of θ .

- (HW-4, #5, and HW-5, #1) Let $\mathbf{X} = (4.3, 0.7, 1.5)$ be a sample of size $n = 3$ from Uniform($0, \theta$) distribution, where θ has Pareto(α, σ) prior. Assume $\alpha = 1$ and $\sigma = 3$.
 - Compute the Bayes estimator of θ under the squared-error loss function.
 - Compute the Bayes estimator of θ under the absolute-error loss function.
 - Compute the generalized maximum likelihood estimator of θ (by looking at the posterior density graph).
 - Find a 95% HPD credible set for the parameter θ .

Hint. For this and other HPD exercises: plot the posterior density of the parameter of interest; looking at it, determine the general form of HPD credible sets; then find the needed quantiles from Monte Carlo simulations.

5. (HW-5, #2) A large shipment of parts is received, out of which five are tested for defects. The number of defects X has $\text{Binomial}(5, \theta)$ distribution. From the history of past shipments, it is known that θ has a $\text{Beta}(1, 9)$ prior distribution. Suppose that $X = 0$ is observed.
 - (a) Find the 95% HPD credible set for θ .
 - (b) Is there a significant evidence that the proportion of defective parts in the whole shipment exceeds 0.1? Conduct a Bayesian test comparing posterior probabilities of the null hypothesis $H_0 : \theta \leq 0.1$ and the alternative hypothesis $H_A : \theta > 0.1$.
 - (c) Revise your conclusion to the previous question taking into account that accepting a large shipment with more than 10% of defective parts is very costly, and thus, the Type II error results in a loss that is 5 times the loss due to the Type I error.
 - (d) Compute the Bayes factor for the test in (b,c).
6. (HW-4, #6) Let $\mathbf{X} = (4.3, 0.7, 1.5)$ be a sample of size $n = 3$ from $\text{Normal}(\theta, \sigma)$ distribution, where σ is known and θ has a non-informative improper prior $\pi(\theta) \equiv 1$. Assume $\sigma = 1$.
 - (a) Compute the Bayes estimator of θ under the squared-error loss function.
 - (b) Compute the Bayes estimator of θ under the absolute-error loss function.
 - (c) Compute the generalized maximum likelihood estimator of θ .

WinBUGS has a constant improper prior `dflat()`. Either use that one or a Normal distribution with a very large variance, say `theta ~ dnorm(0, 0.000001)`, which is a more common approach. Notice that WinBUGS understands the second parameter of Normal distribution as precision, a reciprocal of variance.
7. (HW-5, #3) From path perturbation of a nearby sun, the mass θ of a neutron star is to be determined. Five perturbations quantified as 1.2, 1.6, 1.3, 1.4, and 1.4 are observed. Each observation is (independently) normally distributed with mean θ and assumed standard deviation $\sigma = 0.2$. A priori nothing is known about θ , so the noninformative prior density $\pi(\theta) \equiv 1$ is used. Find a 90% HPD credible set for θ .
8. (Midterm Exam, #3bc) We observe one variable $X = 2$ from the $\text{Normal}(0, \theta)$ distribution, where $\mathbf{E}(X) = 0$ and $\text{Var}(X) = \theta$. One variable is not enough to estimate variance θ , but we also have a prior distribution, which is *Inverse Gamma* (α, β) with $\alpha = 2.5$ and $\beta = 0.5$.
 - (a) Calculate the Bayes estimator of θ under the squared-error loss function.
 - (b) Calculate the posterior variance of your estimator.